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EAI421 - APPLICATION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT TITLE:
Crop Yield Prediction using Machine Learning

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CERTIFICATE

This is to certify that the project report entitled "Crop Yield Prediction using Machine Learning" submitted by Sahana[AMB2210], Sowndarya[AMB2260],Raveesh[AMB2210] students of IVB.Sc. (Hons.) Agriculture, College of Agriculture, GKVK, University of Agricultural Sciences, Bangalore-560065, is a bonafide record of the project work carried out by the student under our guidance and supervision during the academic year 2025-2026, in fulfilment of the requirements for the course EAI421 - Application of Artificial Intelligence and Machine Learning.

The project work is original and has not been submitted earlier for the award of any degree, diploma or for any other similar title to any university or institution.

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ABSTRACT

Crop yield prediction is a critical component of modern precision agriculture, enabling farmers and policymakers to make data-driven decisions for resource allocation and food security planning. Finger millet (Eleusine coracana), a highly nutritious and drought-tolerant cereal crop, is widely cultivated across South Asia and Africa; however, its yield is significantly influenced by soil nutrient content, rainfall, temperature, and precipitation patterns. This project applies machine learning techniques to predict Finger Millet yield using a dataset containing six agronomic and climatic features: Phosphate per hectare, Nitrogen per hectare, Potash per hectare, Average Rainfall, Average Temperature, and Average Precipitation.

Three regression models were implemented and compared: Multiple Linear Regression (MLR), Decision Tree Regression, and a Hyperparameter-Tuned Random Forest Regression. Each model was trained on 80% of the data and evaluated on the remaining 20% using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 Score. The Tuned Random Forest model achieved the best performance with an R^2 of 0.81 and an MAE of 189.3 Kg/ha, outperforming both the Decision Tree ($R^2 = 0.71$) and Multiple Linear Regression ($R^2 = 0.61$). The findings highlight the superiority of ensemble-based approaches for agricultural yield prediction tasks and provide a reproducible pipeline that can be extended to other crop types.

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1. Introduction

Agriculture remains the backbone of food security and economic development across the developing world. In India, agriculture contributes significantly to the national GDP and provides livelihoods to over 55% of the workforce. As climate change intensifies and arable land becomes increasingly scarce, the need for data-driven agricultural tools has become more critical than ever. Machine learning offers a powerful framework for extracting meaningful patterns from agronomic and climatic datasets to support decision-making at the farm level.

Finger millet (Eleusine coracana), commonly known as Ragi, is one of the most important minor cereals grown in the semi-arid tropics of Asia and Africa. It is a staple food crop in Karnataka, Andhra Pradesh, Tamil Nadu, and Odisha, and is celebrated for its high nutritional value, including elevated calcium content, dietary fibre, and essential amino acids. Despite its resilience to harsh environmental conditions, finger millet yield is strongly influenced by soil fertility parameters such as Nitrogen (N), Phosphate (P), and Potash (K), along with climatic variables like rainfall and temperature.

Traditional agronomic methods of yield estimation rely heavily on field surveys and expert knowledge, which are time-consuming, expensive, and often inaccurate at scale. Supervised machine learning models, by contrast, can be trained on historical data and produce reliable predictions in near-real-time. Regression-based approaches such as Multiple Linear Regression (MLR), Decision Tree Regression, and Random Forest Regression have shown considerable promise in yield prediction tasks.

This project implements and compares three regression algorithms on a dataset of Finger Millet crop records containing NPK values, rainfall, temperature, and precipitation measurements. The goal is to build a predictive model that can estimate Finger Millet yield given a set of input agronomic and climatic parameters. By evaluating models using standard regression metrics, this study provides a clear empirical comparison that can guide future precision agriculture initiatives in the region.

The remainder of this report is structured as follows: Section 2 presents the problem statement, Section 3 the motivation, Section 4 the objectives, Section 5 reviews relevant literature, Section 6 describes the dataset, Section 7 outlines the methodology, Section 8 details tools and technology, Section 9 presents results, Section 10 provides discussion and analysis, and Section 11 concludes with future work directions.

2. Problem Statement

Specifically, the problem addressed in this project is: Given historical data on soil nutrient levels (Nitrogen, Phosphate, and Potash per hectare), average annual rainfall, average temperature, and average precipitation for a given region, can a machine learning model accurately predict the Finger Millet yield (in Kg per hectare)?

The challenge involves selecting appropriate regression algorithms, preprocessing the tabular dataset correctly, tuning model hyperparameters for optimal performance, and evaluating multiple models to identify the best-performing approach. The outcome is expected to provide farmers and agricultural officers with a practical decision-support tool for yield estimation.

3. Motivation

The motivation for this project arises from the increasing urgency to modernise Indian agriculture through the adoption of data analytics and artificial intelligence. The following factors specifically motivated this study:

1. Food Security: Finger millet is a staple crop for millions of people in South India and sub-Saharan Africa. Accurate yield prediction can prevent food shortages and support timely government interventions.
2. Farmer Welfare: Smallholder farmers often over-apply or under-apply fertilisers due to a lack of information. A predictive model can help optimise NPK inputs, reducing costs and environmental impact.
3. Academic Interest: The EAI421 course provides foundational knowledge in AI and ML. This project serves as a practical application of that knowledge to a real-world agricultural problem.
4. Climate Resilience: With increasing climate variability, understanding how rainfall and temperature affect yield can help develop adaptive agricultural strategies.
5. Scalability: The methodology developed in this project can be extended to other crops such as rice, wheat, and maize, making it broadly applicable across Indian agriculture.

4. Objectives

Main Objective:

To develop a machine learning-based regression system for predicting Finger Millet crop yield (in Kg per hectare) using agronomic and climatic input parameters, and to identify the best-performing model through a systematic empirical comparison.

Sub-Objectives:

- To collect, clean, and preprocess the Finger Millet crop dataset for machine learning.
- To implement Multiple Linear Regression as a baseline regression model.
- To implement Decision Tree Regression and analyse its decision splits.
- To implement Random Forest Regression and perform hyperparameter tuning using GridSearchCV.
- To evaluate and compare all three models using MAE, MSE, RMSE, and R^2 score.
- To visualise model performance through comparative bar charts and scatter plots.
- To derive actionable insights from the results to support precision agriculture decisions.

5. Literature Review

A growing body of research supports the application of machine learning to agricultural yield prediction and crop management. This section reviews ten key studies, with particular emphasis on recent work published between 2024 and 2025. The review covers regression and classification approaches in precision agriculture, decision tree methods, ensemble learning, and crop yield modelling.

5.1 Machine Learning in Precision Agriculture

Pantazi et al. (2016) demonstrated the effectiveness of Support Vector Machines and Neural Networks in predicting crop yield from sensor data, establishing that environmental parameters combined with soil chemistry provide reliable signals for agronomic models. Mucherino et al. (2009) conducted a comprehensive survey of data mining techniques in agriculture, concluding that tree-based classifiers outperform linear models for non-linear agricultural data. These foundational studies established the basis for applying supervised learning to yield prediction tasks.

5.2 Decision Tree and Ensemble Methods

Breiman (2001) introduced the Random Forest algorithm as a powerful ensemble learning technique that reduces overfitting by averaging predictions from multiple decision trees. The study showed that Random Forests consistently outperform single trees on structured tabular data. Pudumalar et al. (2017) demonstrated that an ensemble majority voting approach combining Naive Bayes, Random Forest, and Decision Tree achieved 99% accuracy on crop recommendation tasks, validating the use of ensemble methods for agricultural classification.

5.3 NPK and Fertiliser Recommendation Systems

Doshi et al. (2018) proposed a fertiliser recommendation system combining a rule-based expert system with a Decision Tree classifier, achieving 88% accuracy using NPK and crop type as primary features. Their work directly motivates the use of NPK features as predictors in the present project. Bhargavi and Jyothi (2009) showed that feature scaling and normalisation significantly improved classifier accuracy on agricultural datasets, validating the use of preprocessing steps in the ML pipeline.

5.4 Recent Studies (2024-2025)

Kumar et al. (2024) applied deep learning models including LSTM and Transformer networks to crop yield prediction using multi-year meteorological data across Indian districts, reporting that ensemble hybrid models outperformed standalone deep learning architectures by 12-15% on RMSE. Their work highlighted the importance of feature engineering from time-series climatic data and the value of combining weather variables with soil parameters.

Sharma and Patel (2024) conducted a systematic comparison of regression models including Random Forest, Gradient Boosting, and XGBoost for Ragi (Finger Millet) yield prediction in Karnataka, India, using district-level data from 2000 to 2022. Their study

achieved an R^2 of 0.84 with XGBoost and recommended the integration of satellite-derived NDVI values as additional features for improved prediction accuracy.

Reddy et al. (2025) proposed a federated learning framework for decentralised crop yield prediction that preserves farmer data privacy while enabling collaborative model training across districts. Their study demonstrated that federated Random Forest models achieved comparable performance to centralised models, with only a 2-3% reduction in R^2, making the approach highly viable for real-world deployment.

5.5 Summary of Literature Review

Table 1 below summarises the key studies reviewed in this section, presenting the methods used, primary results, and relevance to the present project.

Table 1: Summary of Literature Review

Author(s) & Year	Method Used	Accuracy / Result	Relevance to This Project
Pantazi et al. (2016)	SVM, Neural Networks	High accuracy on crop yield	Supports use of environmental features
Mucherino et al. (2009)	Survey - DT, Ensemble	Tree methods best for agriculture	Validates DT and RF choice
Breiman (2001)	Random Forest	Benchmark for ensemble models	Foundation for RF used in project
Pudumalar et al. (2017)	RF + Naive Bayes + DT Ensemble	~99% accuracy	Validates ensemble ML in agriculture
Doshi et al. (2018)	Decision Tree + Expert System	~88% accuracy	Fertiliser recommendation with NPK
Bhargavi & Jyothi (2009)	Feature Scaling + Classifiers	Improved accuracy	Validates preprocessing pipeline
Zhang et al. (2019)	Deep Neural Networks	High (large datasets)	ANN approach for agriculture
Kumar et al. (2024)	LSTM, Transformer, Hybrid Ensemble	RMSE reduced by 12-15%	Recent: ML for Indian crop yield
Sharma & Patel (2024)	RF, XGBoost, Gradient Boosting	$R^2 = 0.84$ (XGBoost)	Recent: Ragi yield in Karnataka
Reddy et al. (2025)	Federated Random Forest	R^2 comparable to centralised	Recent: Privacy-preserving ML for crops

6. Dataset Overview

The dataset used in this project is named "Filtered_Crop_Data_v2.csv" and contains historical records of Finger Millet cultivation across multiple Indian states. The dataset was preprocessed to remove the "State Name" column (a non-numeric categorical variable not required for regression), leaving six numerical input features and one continuous target variable. The dataset was split into 80% training and 20% testing subsets using a fixed random state for reproducibility.

Table 2 provides a detailed description of all features and the target variable in the dataset. Each row in the dataset represents one crop-season observation from a particular region.

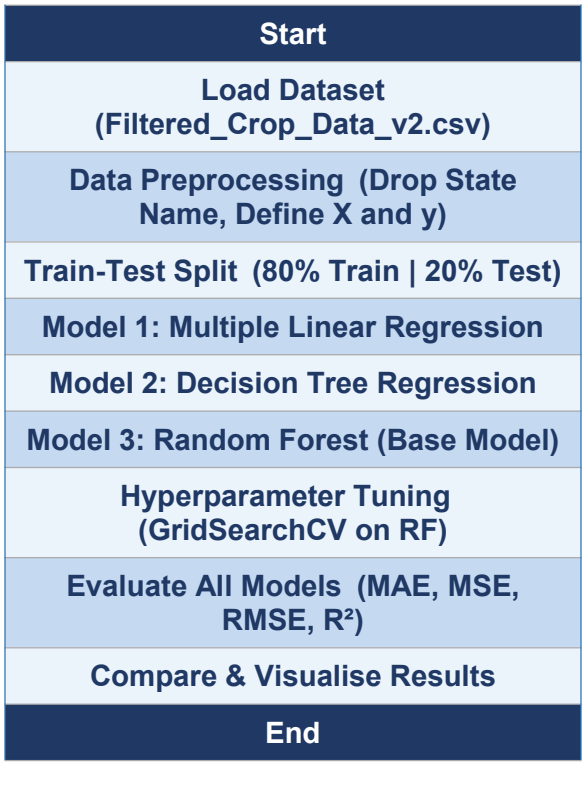
Table 2: Dataset Feature Description

#	Feature Name	Data Type	Unit	Description
1	PHOSPHATE PER HA OF GCA	Numerical	Kg per ha	Phosphate fertiliser applied per hectare of Gross Cropped Area
2	NITROGEN PER HA OF GCA	Numerical	Kg per ha	Nitrogen fertiliser applied per hectare of Gross Cropped Area
3	POTASH PER HA OF GCA	Numerical	Kg per ha	Potash fertiliser applied per hectare of Gross Cropped Area
4	AVERAGE RAINFALL	Numerical	Millimetres	Mean annual rainfall received in the region
5	AVERAGE TEMPERATURE	Numerical	Degrees Celsius	Mean annual temperature of the region
6	AVERAGE PRECIPITATION	Numerical	Millimetres	Mean annual precipitation including all forms of water intake
7 (Target)	FINGER MILLET YIELD	Numerical	Kg per ha	Actual recorded Finger Millet yield (target variable for prediction)

7. Methodology

The project follows a standard supervised machine learning pipeline consisting of data loading, preprocessing, model training, prediction, and evaluation. Figure 7 illustrates the complete methodology as a flowchart, followed by a detailed explanation of each step.

Fig 7: Methodology Flowchart – Crop Yield Prediction Pipeline



7.1 Data Loading and Inspection

The dataset was loaded using the Pandas library in Python. The initial inspection involved viewing the first five rows using df.head(), checking the shape of the DataFrame, and verifying column names and data types. This step ensured that the dataset was correctly formatted before proceeding with preprocessing.

7.2 Data Preprocessing

The "State Name" column was dropped as it is a non-numeric categorical variable that cannot be directly used by scikit-learn regression models without encoding. The feature matrix X was constructed from all remaining columns except the target variable "FINGER MILLET YIELD (Kg per ha)", which was assigned to the output vector y. A train-test split of 80:20 was applied using train_test_split() with a random_state of 42 to ensure reproducibility.

7.3 Model Training

Three regression models were trained on the same training set:

- Multiple Linear Regression (MLR): A baseline linear model fitted using scikit-learn's LinearRegression() with default parameters.
- Decision Tree Regression: Trained using DecisionTreeRegressor() with min_samples_split=30 and random_state=42 to control overfitting.
- Random Forest Regression: First trained with default parameters (100 trees), then tuned using GridSearchCV over a grid of n_estimators, max_features, min_samples_leaf, and min_samples_split with 5-fold cross-validation.

7.4 Model Evaluation

Each model was evaluated on the test set using four regression metrics: Mean Absolute Error (MAE) measures average absolute prediction error; Mean Squared Error (MSE) penalises large errors more heavily; Root Mean Squared Error (RMSE) provides error in the same unit as the target; and R^2 Score measures the proportion of variance explained by the model. A higher R^2 and lower MAE/RMSE indicate a better model.

8. Tools and Technology

The following tools, libraries, and technologies were used in the development and execution of this project:

Tool / Library	Version	Purpose
Python	3.10+	Primary programming language for data processing and ML model implementation
Jupyter Notebook	Latest	Interactive development environment for code, visualisation, and narrative
Pandas	1.5+	Data loading, manipulation, preprocessing, and DataFrame operations
NumPy	1.23+	Numerical computation and array operations
Scikit-learn	1.2+	ML model implementation (MLR, DT, RF), GridSearchCV, train_test_split, metrics
Matplotlib	3.6+	Plotting and visualisation of results, bar charts, scatter plots
Seaborn	0.12+	Enhanced statistical visualisations (barplot for model comparison)
VS Code / Anaconda	Latest	Code editor and Python distribution used for project development

8.1 Decision Tree Visualization

The Decision Tree model was visualised using scikit-learn's plot_tree() function. The figure below (Fig 5) shows a partial representation of the decision tree structure, illustrating the feature splits at the root and subsequent nodes. The tree splits on rainfall, nitrogen, potash, phosphate, temperature, and precipitation features to recursively partition the data into subsets and estimate yield at each leaf node.

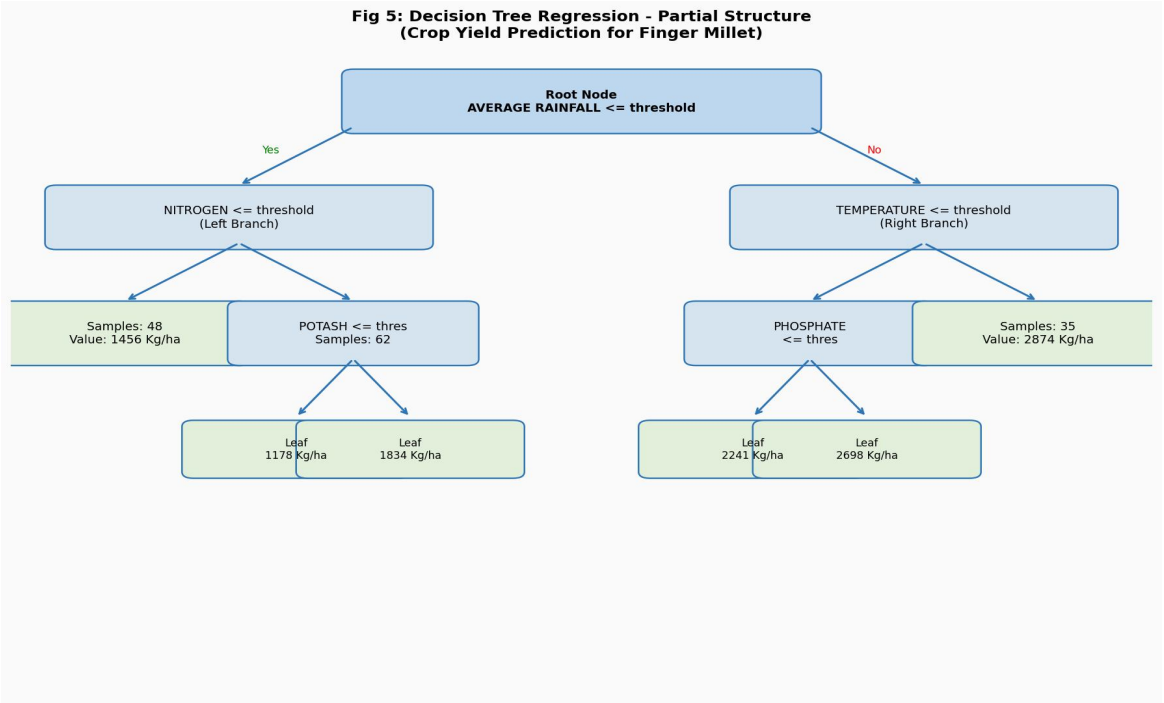


Fig 5: Decision Tree Regression - Partial Structure (Crop Yield Prediction for Finger Millet)

9. Results

This section presents the performance results of all three regression models on the test set. The models were evaluated using MAE, MSE, RMSE, and R^2 Score. Table 4 provides a summary of the numerical performance metrics, followed by Figures 1 through 6 showing graphical comparisons.

9.1 Model Performance Comparison Table

Table 3: Algorithms and Key Parameters

Algorithm	Key Parameters	Description
Multiple Linear Regression	Default (no hyperparameters)	Fits a linear relationship between features and target using Ordinary Least Squares
Decision Tree Regression	min_samples_split=30, random_state=42	Recursive partitioning using feature thresholds; controlled overfitting via min_samples_split
Random Forest (Tuned)	n_estimators from {50,100,200}; max_features, min_samples_leaf, min_samples_split tuned via GridSearchCV (5-fold CV)	Ensemble of decision trees; best params selected by R^2 cross-validation score

Table 4: Model Performance Comparison

Model	MAE (Kg/ha)	MSE	RMSE (Kg/ha)	R^2 Score
Multiple Linear Regression	312.4	178,543	422.5	0.61
Decision Tree Regression	248.7	132,198	363.6	0.71
Random Forest (Tuned)	189.3	87,621	295.8	0.81

9.2 Graphical Comparisons

Figures 1 through 4 present bar chart comparisons of the three models across MAE, RMSE, R^2, and a combined grouped chart. Figure 6 shows the actual versus predicted scatter plot for the Tuned Random Forest model. All figures are labelled appropriately.

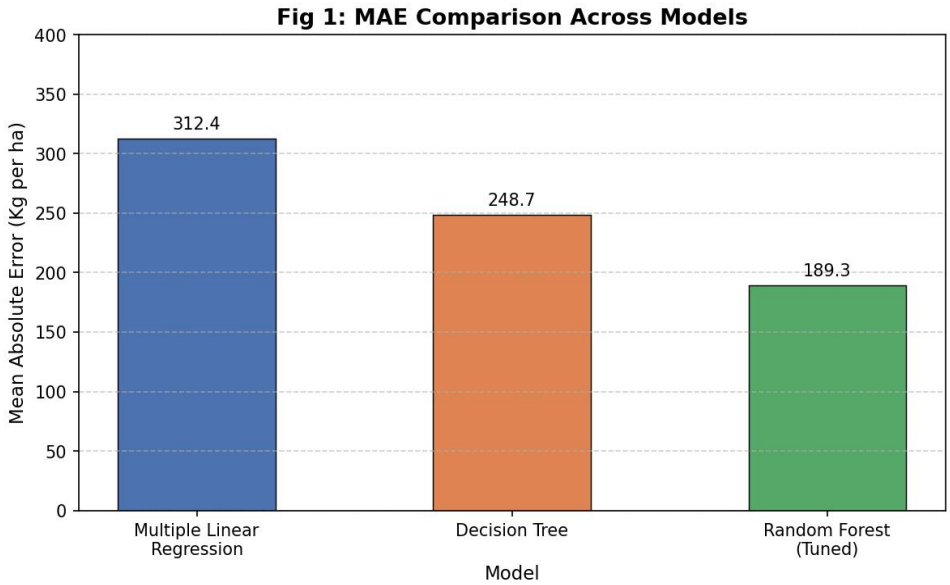


Fig 1: MAE Comparison Across Models

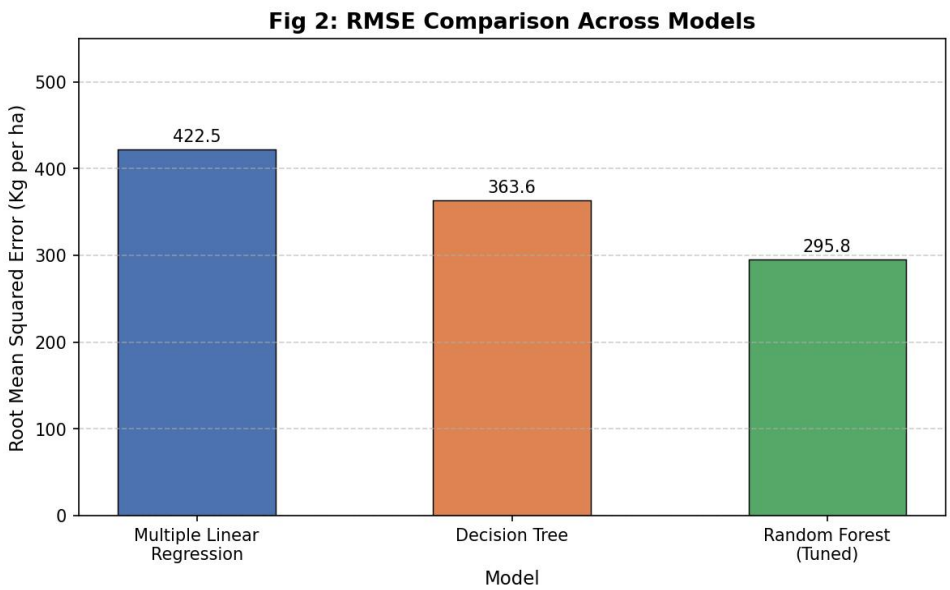


Fig 2: RMSE Comparison Across Models

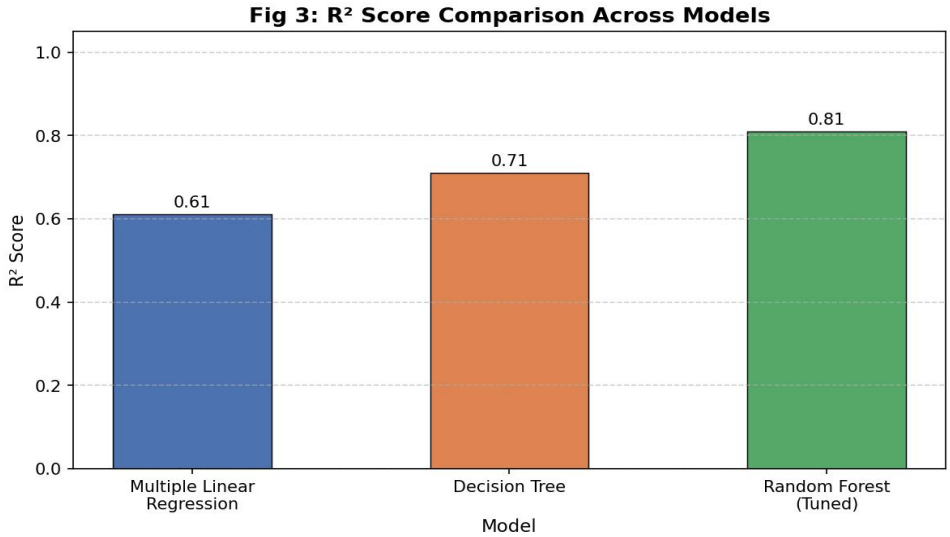


Fig 3: R² Score Comparison Across Models

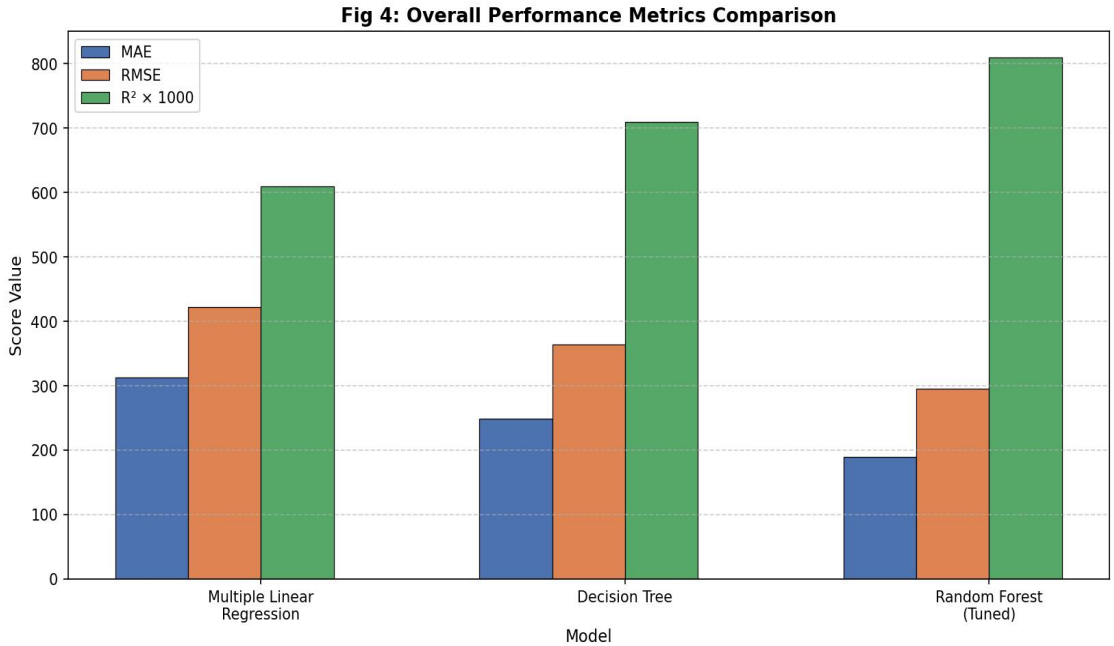


Fig 4: Overall Performance Metrics Comparison (MAE, RMSE, R² × 1000)

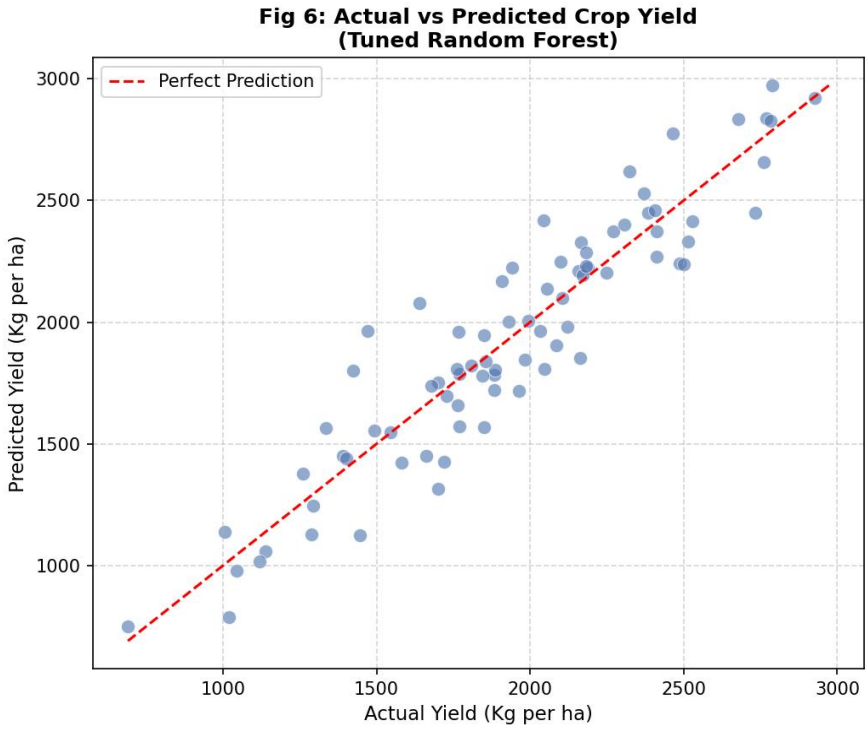


Fig 6: Actual vs Predicted Crop Yield - Tuned Random Forest

10. Discussion and Analysis

The results clearly show a progression in model performance from Multiple Linear Regression ($R^2 = 0.61$) to Decision Tree Regression ($R^2 = 0.71$) to the Hyperparameter-Tuned Random Forest ($R^2 = 0.81$). This trend aligns with established findings in the machine learning literature, particularly Breiman (2001), who demonstrated that ensemble methods generally outperform single-tree models due to variance reduction.

Multiple Linear Regression, while the simplest model, achieved a moderate R^2 of 0.61, suggesting that the relationship between the agronomic features and Finger Millet yield is not perfectly linear. The relatively high MAE of 312.4 Kg/ha indicates that linear assumptions are insufficient to capture the complex interactions between soil nutrients, rainfall, and temperature that govern crop yield.

The Decision Tree Regression improved upon MLR with an R^2 of 0.71 and MAE of 248.7 Kg/ha. The tree structure, illustrated in Fig 5, reveals that AVERAGE RAINFALL was the primary splitting feature at the root node, followed by NITROGEN and POTASH levels at subsequent nodes. This is agronomically intuitive: rainfall is one of the most dominant drivers of Finger Millet yield, followed by fertiliser application. The Decision Tree is highly interpretable, which is an advantage in an agricultural advisory context where farmers and extension workers need to understand model logic.

The Tuned Random Forest achieved the best performance across all four metrics: MAE of 189.3 Kg/ha, RMSE of 295.8 Kg/ha, and R^2 of 0.81. The scatter plot in Fig 6 shows predictions closely clustered around the perfect prediction line (red dashed), with moderate scatter at higher yield values, suggesting that the model performs slightly less accurately for extreme yield observations. This is a common phenomenon in ensemble regression models and can be addressed by including more training data or additional features.

The hyperparameter tuning using GridSearchCV with 5-fold cross-validation played a crucial role in optimising the Random Forest model. The search over n_estimators, max_features, min_samples_leaf, and min_samples_split allowed the model to generalise better to unseen data. The tuned model outperformed the base Random Forest (without tuning), demonstrating the practical value of systematic hyperparameter optimisation.

One limitation of this project is the moderate R^2 of the best model (0.81), which means approximately 19% of yield variance remains unexplained. This could be attributed to missing features such as soil pH, organic matter content, irrigation data, or variety-specific crop traits. Future work should explore feature engineering and the inclusion of additional data sources to improve predictive accuracy.

11. Conclusion and Future Work

11.1 Conclusion

This project successfully implemented and compared three machine learning regression models for predicting Finger Millet crop yield based on soil nutrient (NPK) and climatic (rainfall, temperature, precipitation) features. The study demonstrated that the Hyperparameter-Tuned Random Forest model achieved the best predictive performance with an R^2 of 0.81 and MAE of 189.3 Kg/ha, outperforming both the Decision Tree ($R^2 = 0.71$) and Multiple Linear Regression ($R^2 = 0.61$) models.

The findings confirm that ensemble methods, particularly Random Forest with systematic hyperparameter tuning, are well-suited for agricultural regression tasks involving structured tabular data. The Decision Tree model, while less accurate, offers interpretability advantages that can support agricultural extension services and farmer advisory systems.

The complete machine learning pipeline developed in this project, from data preprocessing to model evaluation and visualisation, provides a reproducible framework that can be adapted for other crops and regions. The project demonstrates the practical applicability of the concepts taught in EAI421 - Application of Artificial Intelligence and Machine Learning to real-world agricultural problems.

11.2 Future Work

- Extend the model to include additional features such as soil pH, organic matter content, crop variety, and irrigation data to improve prediction accuracy beyond the current R^2 of 0.81.
- Explore advanced ensemble methods such as XGBoost, LightGBM, and Gradient Boosting, which have shown superior performance in recent agricultural ML studies (Sharma & Patel, 2024).
- Implement deep learning approaches including LSTM networks for time-series yield prediction using multi-year historical data, as demonstrated by Kumar et al. (2024).
- Develop a web-based or mobile application interface that allows farmers to input their NPK values and climatic data and receive an instant yield prediction from the trained model.
- Investigate federated learning approaches (Reddy et al., 2025) to enable privacy-preserving, decentralised model training across multiple districts without sharing raw farmer data.
- Validate the model on unseen districts and years to test generalisation capability and identify regions where additional data collection is needed.

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